



**Decentralized identity authentication using blockchain and
machine learning-based fraud detection**

*MAJOR PROJECT REPORT submitted in partial fulfillment of the requirements
for the Award of the Degree of*

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(AUTONOMOUS - AFFILIATED TO JNTU-K, KAKINADA)**

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CERTIFICATE

This is to certify that this project report titled “**Decentralized identity authentication using blockchain and machine learning-based fraud detection**” is a Bonafide record of work done by **SURESH CHITTIBOINA (228W1A1279)** and **GANESH SANKAR KUMAR KARNAM (228W1A1295)** under my guidance and supervision is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Technology, **V.R. Siddhartha Engineering College** (Autonomous under JNTUK) during the year 2024-25.

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PROJECT SUMMARY

S.No	Item	Description
1	Project Title	Integrating Blockchain And Machine Learning For Identity Verification And Fraud Detection
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3	Name of The Guide	Dr.S.Jaya Prakash
4	Research Group	Blockchain and Machine Learnig
5	Application Area	Financial Sector
6	Aim of the Project	The aim of the project is to develop a decentralized and secure user authentication system by integrating blockchain technology and machine learning.
7	Project Outcomes	The outcome of the project is a fully functional decentralized application (DApp) that allows users to securely register and log in using blockchain technology.

Student Signatures

- 1.
- 2.

Signature of the Guide

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ABSTRACT

Today, financial fraud and identity theft have become rather frequent problems as financial transactions and payments occur in a digital environment. Conventional fraud detection techniques usually use a centralized approach, hence susceptible to security attacks. Thus, this project proposes an innovative technique to detect fraud and verify identities by integrating the capabilities of blockchain and machine learning (ML). Today, in the digital era, financial fraud and identity theft have become rather frequent problems as financial transactions and payments occur in a digital environment. Conventional fraud detection techniques usually use a centralized approach, hence susceptible to security attacks. In this regard, this project proposes an effective technique to detect fraud and verify identities by integrating the capabilities of blockchain and machine learning (ML) algorithms. Blockchain assures integrity, transparency and immutability of identity information and transaction details.

Keywords: *Blockchain, Machine Learning, Identity Verification, Fraud Detection, Ethereum, Smart Contracts, Random Forest, Ganache, MetaMask, Decentralized Application*

CHAPTER-1

INTRODUCTION

However, the rise in number of online purchases and payment transactions has raised the probability of frauds as well. Traditional payment methods are dependent on centralized databases and hence can be easily accessed by malicious entities, thus making them less secure and reliable. Thus, there is a requirement for building a smarter payment system that will be not only reliable but also will be able to detect fraudulent activities. Blockchains provide a way to store user data in a safe and tamperproof manner. Blockchain technology ensures trustworthiness and transparency. However, blockchains do not have mechanisms for data analysis and identifying frauds. Hence, we will be using machine learning algorithms for fraud detection in the proposed project. ML models have been shown to analyze transaction data and detect frauds. Multiple models will be trained to select the most efficient one. Models including XGBoost, LightGBM, Random Forest, Gradient Boosting, and Logistic regression models will be trained. Afterward, their performance on a test dataset will be analyzed. The output of ML algorithms is often in terms of probability of a specific class being true or false. Decision making based on probabilities is hard for the end-user. Thus, threshold decision engine will convert probability value to a binary decision of being legitimate or fraudulent transaction. The proposed system includes distributed file storage via IPFS in which data is securely stored by encrypting sensitive information. Only hashes of uploaded files are saved in blockchain, thus providing extra security and speed benefits to the system. The system also features real-time transaction dashboard in which transaction details can be viewed by administrators and end-users. The dashboard can alert about suspicious activity and show the probability of frauds. Finally, administrative controls will allow identifying suspicious users. Overall, the project aims at combining the capabilities of machine learning, blockchain, and IPFS in order to build a safer payment system.

1.1 Origin of the Problem

Because of the development of the digital banking, e-commerce, and digital payments market segments, fraud has been recognized as an important problem. The majority of current financial services have a centralized nature and are based on the centralized

database management system. Centralized approaches to data storage are less protected than distributed systems due to the increased risks related to data leaks, hacking, and data modifications that lead to identity theft and monetary losses. Furthermore, modern systems for detecting fraud activities are rather rigid as they follow certain rules and are not able to modify their functioning based on new threats and challenges. Static systems fail to handle new types of attacks as they do not adapt and recognize any changes. Finally, one should consider the fact that the amount of daily transactions has dramatically grown. There is an increased need in the automation of fraud detection and transaction analysis. Blockchain offers certain improvements in the field due to its unique approach to the organization of data storage. Nevertheless, Blockchain does not allow to analyze transaction data and detect fraud. In turn, machine learning algorithms provide efficient solutions for identifying and analyzing potentially dangerous activity. However, ML tools do not offer security guarantees and are transparent. One can conclude that there is a high necessity for using the combined approaches involving Blockchain, machine learning, and other tools.

1.2 Basic definitions and Background

1.2.1 Blockchain

Blockchain is a decentralized digital system that securely and transparently stores data. Instead of keeping data in one central location, it is stored across many devices connected in a network. Information is grouped into blocks, and these blocks are connected in a sequence, forming a chain. A major advantage of blockchain is its immutability. Once data is added to a block, it cannot be altered or removed. This makes it highly secure for keeping important information such as transaction records or user details. In this project, blockchain is used for user authentication and to securely store information related to fraud through the use of smart contracts.

1.2.2 Smart Contract

Smart contracts are self-executing contracts with the terms of the agreement between buyer and seller are directly written into lines of code. They automatically enforce and/or perform contractual clauses when predetermined conditions are met. With these contracts, we won't need intermediary hence all transactions are carried out securely and transparently. Users registration, login verification and fraud status update will be achieved through Smart contracts in the system.

1.2.3 IPFS

IPFS stands for Interplanetary File System, is a decentralized storage system that is used to store files across a distributed network. Unlike traditional systems which stores data in just a single location, IPFS breaks the file into blocks and spreading these blocks across different nodes in the network. In this project, important files are encrypted and saved in IPFS. On the blockchain only the hash of file (CID) is stored hence it saves storage and adds to the security.

1.2.4 Fraud Detection

Fraud detection refers to the process of using an algorithm, which is constantly learning from the data it processes, to automatically determine whether a fraudulent activity has taken place or not. In finance this often involves analyzing transaction history in order to identify any unusual behaviour that might occur. Machine learning is a driving factor in the fight against cyber fraud.

1.2.5 Machine Learning

Machine learning is a form of intelligence system that allows machines to learn data and makes inferences without telling what the exact output is. It can find patterns from the data with the help of some rules. As fraud detection is a type of machine learning model, machine learning models use transaction data to find the anomaly patterns which shows the system is trying to commit fraud. We utilize many different kinds of machine learning models to find the fraud in our project including XGBoost, LightGBM, Random Forest, Gradient Boosting and Logistic Regression. In this project we are focusing on using machine learning models for fraud detection and it is excellent at doing that task.

1.2.6 Decision engine

Decision engine: decision engine is the block in the system which considers the opinion of the machine learning models and then makes the decision using some rules. In this project the decision engine takes a threshold value to declare the transaction as fraud or non-fraud depending on the predicted probability for fraud transaction. It declares a transaction as fraud if predicted probability is greater than the threshold. It declares transaction as fraud if predicted probability is less than threshold.

1.3 Problem Statement

The quantity of transactions is currently increasing in an alarming rate, leading to major financial fraud and identity theft issue. Present system of fraud detection architecture uses single database which stores the whole transaction history and user data and this type of systems are not secured as these are prone to security attacks, manipulation of data and unauthorized access. Old detection methods rely heavily on rule based system, and thus are not proficient in detection of new and complex fraud patterns. The conventional method cannot handle huge volume of transaction data and often misses the fraudulent transactions. ML algorithms can predict the transactions but does not assure security and safety of data, whereas the blockchain is able to secure and maintain the data but cannot analyze the transaction to find out fraud. So it is very important to design a secure, and an intelligent system. Our main objective of this problem is to design a decentralized and efficient fraud detection system using machine learning and blockchain. The designed system will able to find the fraudulent transaction, ensure data security and enable decision-making in real time. It will use fraud detection, and blockchain technology. The importance of fraud detection cannot be over stated; the system relies on blockchain technology for a secure system.

1.3.1 Objectives of the Study

The main purpose of this research effort is to create and build an effective and secure method for identifying fraud in financial transactions using state-of-the-Art technology. By utilizing machine learning, as well as comparing and selecting the machine learning algorithm that performs the best, the goal is increase accuracy of fraud detection. In addition to improving accuracy, another goal of this study is to provide security and transparency to the data being utilised to detect fraud by leveraging blockchain-based technologies. Using blockchain will help safeguard customer information stored by Financial Institutions and will allow for a higher level of trust from customers since the data stored on a blockchain cannot be altered after the fact. As mentioned previously, this research project is also looking toward developing a decision-making process for determining whether a transaction is either fraudulent or legitimate based upon pre-defined criteria. This will improve the dependability of this type of system. Lastly, the goal of this study is to develop a complete system that integrates a machine learning based fraud detection system with a blockchain and visualisation technologies into one

concise system that conforms to both security and practicality for use in today's society.

1.3.2 Expected Outcomes

The goal of this research study is to develop a system that can accurately detect fraudulent transactions by utilizing many various ML algorithms. A key objective of the research will be to build a model that will determine which algorithms are most effective for creating an ML model to detect fraud in different scenarios. Additionally, it is expected that the system will ensure user data integrity and prevent data tampering or modification via the use of block chain technology through decentralization of data. Implementing a decision making mechanism within the system is expected to provide a better classification of transactions, which will then help to improve the performance of the system by reducing uncertainty. Furthermore, it is expected that the proposed project will provide a visual dashboard that will allow end users to monitor transaction activity and review suspected fraudulent activities in real time. This function can enhance the user's ability to evaluate and interpret the overall system in general and quickly detect suspicious activity within the system. Finally, the end result will be a complete and integrated system containing components related to ML, block chain technology and user interface elements, so that it can be used practically and enhanced in the future.

1.4 Applications

The proposed system is really useful in areas where we need to make sure transactions are secure and fraud does not happen. For example in banking and financial services it helps to find transactions that are not authorized credit card fraud and things that are suspicious with accounts. E-commerce platforms can use the proposed system to find out if orders are fake and if there is payment fraud so users can have transactions.

The proposed system is also useful for payment systems and mobile wallets because it can detect fraud in real time and make sure transactions are secure. We can use the proposed system to verify identities, which helps to prevent people from stealing identities. It uses blockchain to authenticate. In the insurance sector the proposed system helps to find claims that're fraudulent by looking at patterns of transactions. Government organizations can use the proposed system to reduce fraud in services, such as subsidies and tax systems. Also the proposed system can be part of cybersecurity systems to monitor activities that're suspicious and make security better overall and the dashboard

that shows things in real time helps organizations to look at trends in transactions and make better decisions, about the proposed system and its uses.

1.4.1 Financial Sector

In order to protect customer information and ensure secure transactions in the financial services industry, detecting fraud is extremely important. Financial institutions are targeted frequently due to their high volume of daily transactional activity wherein an individual can commit a crime such as identity theft, create a fictitious commercial transaction, or launder money (i.e. concealed illegal activity). The system we are discussing allows companies to observe real-time transactions for evidence of fraudulent behaviour via pattern recognition. This method uses machine learning algorithms that can analyze historical transaction databases and detect outliers compared to future/new/real time transactions. The utilization of blockchain (distributed ledger) architecture provides an immutable and irreversible record of all transactions which will ultimately drive up trust in the system, as it will become significantly more difficult for end-users to manipulate the transaction data. Our solution represents an enhancement of both the detection of fraud and the security of transactions, as well as providing ways for financial services firms to monitor and mitigate risk to their customers effectively and efficiently.

1.4.2 Cybersecurity Systems

Detecting suspicious events and unauthorized access are prominent issues in security systems. With growing number of digital services and online platforms, threats to cyber space like hacking, phishing, identity misuse are on the rise. The pre-sented method may also be used in cyber security to monitor the behavior of a user and detect anomalies in the behavior. ML models are utilized to analyze the data and identify threats or potential threats as they detect abnormal usage that is not consistent with regular patterns of use. The implementation of blockchain technology provides an additional security layer as the user data and the system logs are saved in a secure, immutable manner. This does not allow attackers to modify sensitive data and enhances system dependability. In addition, the system administrator can monitor system activities in real-time with a monitoring dashboard and be notified immediately of any suspicious activities. This enhances the overall efficiency of the cyber security and assists in securing the digital environment.

1.4.3 E-Commerce Platforms

On e-commerce sites, many transactions happen every day, so they are a big target for scams such as fake orders, payment fraud and account takeovers. Security of the transaction and customer information are two major issues that are difficult to deal with in online shopping. Our approach is general and also enables online detection of suspicious transactions in e-commerce transactions. ML models are trained to recognize the transaction patterns and flag if there are any anomalies that can be fraud. This not only prevents fraudulent purchases, but also protects the sellers and customers involved. The technology of blockchain provides a safer, more open and incorruptible transaction record. This helps to limit the potential for data manipulation and increases trust. Further, the adoption of decentralized storage makes it possible to treat sensitive data in way that is compliant. The real-time dashboard provides administrators with the tools to monitor transactions, identify evolving fraud patterns and respond in real time to suspicious activity. Overall, the framework increases transaction security, minimizes financial losses, and bolsters the trust of e-commerce platform users.

1.5 Summary

In this part of the book we talk about the ideas and background of the system we are proposing. We need a system that can detect fraud in an efficient way. The old ways of doing things have some problems so we explain why we need something. We also talk about some technologies like machine learning, blockchain, smart contracts and IPFS so we can understand the parts of the system. We say what the problem is and what we want to achieve with this study. We look at areas where this system can be used such as banks, cybersecurity systems and online shopping platforms. This shows us how the system can be used in life. This part of the book gives us an understanding of the system we are proposing and why it is important. It gets us ready for the parts of the book, where we talk about how we did the work how we made the system and what we found out. We will discuss the system and the results in detail in the following chapters, about the system.

CHAPTER-2

REVIEW OF LITERATURE

This chapter describes the review of literature that we have taken from various paper sand considered all the points mentioned in the papers.

2.1 Description of Existing Systems

Table 2.1.1 : Literature review

SNO	Title & Publication Details	Key Focus	Relevance to Project	Findings and contributions
1	Data Mining for Credit Card Fraud: A Comparative Study (2011) S. Bhattacharyya et al., <i>Decision Support Systems</i>	Comparison of machine learning models such as Logistic Regression and Random Forest for fraud detection	Helps in understanding basic ML models and their performance in fraud detection systems	Shows that machine learning models outperform rule-based systems, but highlights challenges like imbalanced datasets and need for better preprocessing
2	XGBoost: A Scalable Tree Boosting System (2016) T. Chen, C. Guestrin, <i>ACM SIGKDD</i>	Development of a scalable and efficient boosting algorithm	Used in the project for high-performance fraud detection	Provides improved accuracy, regularization, and scalability; performs well on large datasets and complex patterns
3	LightGBM: A Highly Efficient Gradient Boosting Decision Tree (2017) G. Ke et al., <i>NeurIPS</i>	Fast and efficient gradient boosting model with low memory usage	Helps in improving speed and efficiency of fraud detection models	Achieves faster training and higher accuracy using leaf-wise tree growth; suitable for large-scale applications
4	Blockchains and Smart Contracts for the Internet of Things (2016) K. Christidis, M.	Use of blockchain and smart contracts for secure	Used for secure user authentication and data storage in the project.	Highlights benefits like immutability, transparency, and trust; also discusses challenges like

	Devetsikiotis, <i>IEEE Access</i>	and decentraliz ed systems		scalability and integration
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Data Mining for Credit Card Fraud: A Comparative Study

Authors: S. Bhattacharyya et al.

Year of Publising:2011

Observations:

The paper provides a thorough comparison of several data mining techniques for credit card fraud detection. Models such as Logistic Regression, Decision Tree and Random Forest are studied and it is demonstrated that the ML models significantly outperform the classical rule based systems. In particular, it is observed that ensemble models such as Random Forest achieve higher prediction accuracy and reduce false positives. Issues such as unbalanced datasets where fraudulent transactions are a tiny minority are addressed and it is reiterated that improved feature selection and pre-processing are required for effective real-world systems.

XGBoost: A Scalable Tree Boosting System

Authors: T. Chen, C. Guestrin

Year of Publising:2016

Observations:

This paper is about XGBoost. XGBoost is a good machine learning algorithm. It is based on something called gradient boosting. XGBoost makes predictions. It does this by fixing the mistakes of models. It fixes these mistakes one at a time. The people who did this study found out some things that make XGBoost very good. These things are regularization and parallel processing and handling missing data. XGBoost is very good with datasets. It is also very good, with problems. So we can use XGBoost to find fraud. The paper also shows that XGBoost can be very accurate. It does not use much computing power.. Xgboost needs the right settings to work really well. We have to tune the parameters of XGBoost right. This way XGBoost will work at its best. XGBoost is a good algorithm.

LightGBM: A Highly Efficient Gradient Boosting Decision Tree

Authors: G. Ke et al.

Year of publising: 2017

Observations:

LightGBM is a way to do gradient boosting that is faster and uses less memory. It grows trees in a way from the leaves, which makes it more accurate than the old way of growing trees level by level. This makes LightGBM really good at handling amounts of data and lots of features. LightGBM is especially good for things that need to happen like finding fraud in real time. The people who made LightGBM say that if you do not set it up right it can make mistakes especially with smaller sets of data. LightGBM can be bad at datasets because it can get too good at guessing the answers, for that specific set of data, which is called overfitting.

Blockchains and Smart Contracts for the Internet of Things

Authors: K. Christidis, M. Devetsikiotis

Year of Publishing: 2016

Observations:

In this paper we talk about how to use smart contracts and blockchain to develop safe systems without any central authority. Data on the blockchain is saved on different computers so that data can't be changed. Trust builds on the transparency on the blockchain as everybody can read data on the blockchain. By removing middlemen we get smart contracts which helps to make transactions on the blockchain. This study found that by using the blockchain data can be kept safe which also helps to prevent and detect frauds. There is also few disadvantages using blockchain, like that data can only be saved after mining process. Processing of data also require lot of power and they can't be easily implemented into existing systems. Using the blockchain together with technology will make blockchain much better for use in real-life scenarios. With use of smart contracts on blockchain the blockchain system gets even more trustworthy for security purposes, where smart contracts will make it possible to write code for transactions on blockchain to run automatically.

2.2 Summary of Literature Study

[1] Data Mining for Credit Card Fraud: A Comparative Study

From literature study, it can be seen that various people had done great research on the aspect of fraud detection with machine learning ways and security approaches. In the past most of research were conducted to explore machine learning algorithms and data mining models to discover deals, it should be mentioned an important work of Bhattacharyya et al. In 2011 which used Logistic Regression and Decision Tree and Random Forest algorithms on credit card fraud detection. According to this research, machine learning models is significantly superior over rule based systems for fraud detection with accuracy; ensemble approaches such as Random Forest which are by minimizing alarm and making stable predictions were considered as best model for detecting fraud. For the recent studies on fraud detection, boosting algorithms such as XGBoost and LightGBM are widely used as they offer better accuracy, efficiency and scalability compared to traditional models; hence these algorithms are more suited for a modern financial fraud detection system as they can handle patterns in data and large amounts of data. Besides this, there were studies on block-chain based systems to ensure data security and transparency, real time detection systems which focused on identifying fraud as deals take place. With the advancements made by various researches in the area of financial fraud detection most research concentrated on either of one area, but there was no system where financial fraud detection and secure storage of data and real-time monitoring were combined. The proposed system aims to cover this gap by combining machine learning models with blockchain technology and real-time dashboard to provide a practical and efficient financial fraud detection system. The proposed system uses fraud detection and block-chain technology to create a financial fraud detection system. Financial fraud detection is a section of the proposed system.

[2] XGBoost: A Scalable Tree Boosting System

As a result, there are still gaps in decision-making, monitoring, security, data safety, and transparency. Several researchers have opted to focus on machine learning

algorithms to analyze transactions to detect fraud. They have employed various algorithms such as Logistic Regression, Random Forest, XGBoost, LightGBM, etc. to make it more accurate. These studies only consider the accuracy of the fraud detection, which makes it incomplete. With the advancement of technology, there's a rise in security breaches and data tampering. Due to this, there's a need for data safety and transparency. Blockchain decentralized storage can result in immutability and trust. Blockchain is majorly used only for storage, which means it will not be helpful unless machine learning is done with blockchain to detect fraud. Data stream processing should be used to detect fraud in a timely manner. Existing systems have weak monitoring, visualization, and user-friendly interfaces. There is a need for an integrated detection, security, monitoring, and decision-making system, thereby completing the fraud detection system. There's a gap in the existing systems where machine learning, blockchain, and real-time analytics are kept separate and they are not integrating the techniques and methods to implement a complete fraud detection system. However, the proposed system would provide a practical solution to all these shortcomings, which will be advantageous for fraud detection models and techniques.

[3] LightGBM: A Highly Efficient Gradient Boosting Decision Tree

Fraudulent transactions are increased in the banking sector, therefore to tackle this problem the concept of machine learning has been introduced. This paper aims at combining the technologies of machine learning to the blockchain. In this machine learning models are developed which helps detect the fraudulent transactions. The outputs obtained from the machine learning models are being given to the decision engine to perform the task of mapping the outputs of the model to fraudulent or not fraudulent. Not only that the information regarding the user data and the information related to the fraud are stored in the blockchain. The concept of blockchain deals with the decentralized handling of the data. In order that the systems are secure, the concept of the smart contracts is used in the blockchain so that there is an access control to the data presented. The activity of the transaction and the trends of the frauds can be seen in the visualized format. The machine learning models are developed on the financial transaction data using models like XGBoost, LightGBM, Random Forest, Gradient Boosting, Logistic Regression. The model performances are also compared with the different parameters to choose the best model to detect the fraud. The framework is real time and provides a complete implementation of how the frauds can be detected and

the data can be securely stored so that a safe and efficient transaction can be processed.

[4] Blockchains and Smart Contracts for the Internet of Things

The project aims to implement a fraud detection system for financial transaction data that uses machine learning models to analyse the transaction data of the users of financial organisations like banks, credit unions, etc. The objective is to identify suspicious transaction behaviours and reducing the class imbalance ratio in the application of the financial sector using machine learning in order to get the best models by tuning the hyper-parameters of the different models-xgboost, LightGBM, Random Forest, Gradient Boosting, Logistic Regression etc. initially trained on the transaction data of the users of the model. The project also used blockchain in order to handle certain data that is collected on the Smart Contract, later stored on the IPFS. The user of the model will be onboarded in a simulated real world like environment while the data generated will be a simulation of user behaviour on the real time dashboard built on streamlit in order to monitor the transactions on whether they are fraudulent or not, provide the statistics of the fraudulent transactions as well as alerts for any suspicious users. In order to allow user login so that only the authorized user could deploy and access the model, used the concept of blockchain wherein who so ever logs in to the model is check right from the individual level upto the organisational level to have a secured authentication and for secured access of the data. The objective of using Blockchain for the above purpose is to maintain the integrity of the data being used and processed by the model for the predictions. Thus The system replaces the conventional database systems with tracking limited to the User Authentication and data storage on the Smart Contract that was instantiated on the blockchain network. The project implements the concept of blockchain on a controlled environment where it is built on a local blockchain network setup and connected to the applications and thus is not deployed on a public blockchain due to which the blockchain has no large scale production data.

2.2 Scope of the Project

The purpose of this paper is to establish a secure and efficient fraud detection system by integrating machine learning techniques with blockchain technology. It processes financial transaction data and detects fraudulent transactions based on learned patterns. The entire project involves building and training multiple ml models such as logistic

regression, random forest, gradient boosting, xgboost and lightgbm. These models are built, trained and tested for the best one to be used for detecting fraud. It also performs data preprocessing such as cleaning, normalization, balancing to enhance model performance. While the fraud detection, the project integrates the blockchain for secure user identification and the storing of critical information. User-centric operations are handled by smart contracts, user data is stored decentrally using IPFS. The system also has a real-time monitoring dashboard, where transaction data, fraud statistics and notifications can be viewed. This enables both users and administrators to track system performance and to respond to suspicious behavior. But the scope is limited to a private environment on a local blockchain network e.g. Ganache. It does not include deployment on a public blockchain and/or processing with a large-volume real-time production data. Future work may involve generalizing the system to accommodate more functionalities and large-scale services.

2.3 Limitations of Existing System

There are several challenges with the existing fraud detection mechanisms which reduce their efficacy when applied to practical situations. Most conventional protocols are based on centralized systems, in which all the data is kept at one location. They are, therefore, susceptible to security attacks, data modification, and illicit use. Several rule-based systems exist, these systems follow a series of predefined rules for the identification of the fraud. These techniques are rigid and cannot accommodate new and changing types of fraud. They miss nonlinear patterns or produce a high number of false positives. Another limitation is the use of a single machine learning model in isolation, without any evaluation/analysis or comparison. This reduces the overall accuracy of the system as the results may vary with the model and the datasets. Also, most of the pre-existing ones handle poorly the class imbalance problem. Particularly since fraudulent activities are extremely scarce when compared to normal ones, models could be biased to predict them as non-fraudulent. No access to real-time tracking is yet another problems. A large portion of these systems do batch processing of data, which means that they have a latency and cannot deliver results immediately as this impedes detection of the fraud. The lack of proper visualization tools also makes it challenging for users to gain insight from system and transaction behaviors. As a final note - from what I understand of the paper, most of the system cannot have sophisticated security measure such as blockchain or decentralised storage either. This creates

opacity in the data and exposes it to potential tampering.

2.4 Challenges of Existing System

Performance and reliability are negatively impacted by a number of challenges that existing common fraud detection mechanisms have to contend with. One of the biggest bottlenecks is processing the transaction data in massive amount. As digital transactions grow in number, systems are hard pressed to process and analyze data quickly enough. There is also the challenge of extremely imbalanced data sets. In general, fraudulent transactions are much fewer than genuine ones. This bias hinders the ability of machine learning models to accurately predict fraud without incurring in false predictions. Keeping up with new and emerging fraud trends is also a big challenge. Fraudsters keep updating their methodologies, and legacy systems have very limited capabilities to identify these new trends. This necessitates systems that can learn and evolve. Security is yet another challenge. Centralized systems are more susceptible to cyberattacks, potential data breaches and unauthorized changes. A critical issue in current systems is ensuring data purity, while storing sensitive information. Also, many systems inhibit efficient integration of their various components including the data processing, detection of fraud, and the surveillance of fraud. This results in duplicated efforts and causes the system to be unmanageable. Lastly, the detection of fraud is known to be theoretically weakened without real time monitoring and alerting mechanisms. Late detection would bring financial losses and undermining of the system reliability.

2.5 Research Gap

Based on the current systems and literature, the solution implementations in fraud detection are more tend to focusing on individual parts of the system rather than the whole integrated system. A great many research have been conducted to optimize machine learning models for better accuracy; but they fail to consider data security and transparency. Some systems employ blockchains to improve security, but they mainly reside in data storage, and do not integrate with intelligent fraud detection techniques. Likewise real-time systems are developed, but lack suitable monitoring, visualization and interactions facilities. Another missing piece is the comparison among different machine learning models. A large number of systems are designed with a single model for all data types. This leads to the decreased general strength and flexibility of the

system. Furthermore, there is often no a priori clear decision rule on how to read prediction results in the existing work. Absent such a rule, one cannot systematically decide which transactions to classify as “good” or “bad.” Hence, there is a challenge to build a framework that can combine advanced machine learning models, secure data management leveraging blockchain technology, and real-time monitoring. The proposed project will focus on bridging these gaps and improve performance as well as usability by employing multi-models, a decision engine and a real-time dashboard.

2.6 Proposed System

In this paper, we present [name], a secure and intelligent fraud detection system based on the collaborative work of machine learning, blockchain and real-time monitoring. Rather than traditional method of using single technology, the proposed system incorporates multiple technologies to drives better accuracy and security. The system utilizes different machine learning algorithms(K-Nearest Neighbor(Logistic Regression, Random Forest, Gradient Boosting, XGBoost and LightGBM) along with transaction data. These models are trained and tested for selection of best fitting model for detecting the fraud. This multi-model methodology enhances reliability and contributes to superior results. A decision engine is added to improve classification intelligibility. A threshold, however, operates on the probability values rather than on the model outputs directly, and a transaction is classified as fraudulent or normal based on this threshold. This assists in consistent and interpretable decisions. A data management system based on blockchain is proposed. Smart contracts manage user authentication and record significant transactions , preventing unauthorized alterations. Also, IPFS is used for ddecentralized storage, where only file hashes are saved in the blockchain, enhancing efficiency. In addition, the system provides real-time monitoring with a dashboard to show transaction information, fraud statistics and warning messages. This enables the user to track activity and respond promptly to potential threats. As a whole, our system offers a fully integrated, secure and high-performance solution for the fraud detection, which can be exploited in real faces.

2.7 Advantages of Proposed System

By integrating machine learning,blockchain and real-time monitoring, our system has the advantages on ease of use, effectiveness and efficiency compared with other similar

fraud detection systems. One of the biggest benefits is an increase in fraud detection accuracy. By combining multiple classifiers and selecting the one that gives the best performance, the final results given by the system are more reliable when compared to the cases of single classifiers. Another significant benefit is the security of data is greater. The use of blockchain means that all records are maintained in a decentralized, tamper-resistant manner, which keeps would-be manipulators at bay and boosts trust in the system. Usage of IPFS as the decentralized storage, reducing the reliance on centralized servers and improving data integrity. It also contributes to efficient storage by having only the file hash in blockchain instead of a file. The system also supports real-time monitoring via an active dashboard. This enables consumers to monitor their transactions, observe fraud trends, and respond to suspicious activities in a timely manner. Also, the decision engine adds to the transparency by turning probability outputs to explicit classes. It helps in better prediction and reduces the uncertainty in prediction. The system also features alerts and a transactional log, enabling even more comprehensive monitoring and analysis. In general, proposed system is more secure, efficient and easy to operate, which can be well applied to the real-world fraud detection practice.

2.8 Summary

In this chapter we review the prior art and related work in fraud detection. The drawbacks and challenges of existing methodologies are presented, which motivate us to design a more secure and efficient scheme. This literature review gives an insight into various machine learning models and blockchain based systems for fraud detection. The gap in the research is evident to be filled due to most of the existing systems being for individual components e.g. machine learning or security and not the fusion of these components. With these considerations in mind, the proposed system is presented as a refined solution. The proposed solution leverages ensemble models and decision engine with blockchain technology and live monitoring to address the challenges of some existing solutions. The merits in terms of higher accuracy, better security, more user-friendly neural network layer etc. are also presented. In conclusion, in this chapter, the motivation for the proposed system was well understood and it leads to the next chapter where the design of the system and implementation are elaborated.

Chapter -3

PROPOSED METHODOLOGY

3.1 Design Methodology

This paper offers a system to secure, decentralize and provide a robust solution to verify users' identity and detect fraud. This system is based on combining Blockchain and Machine Learning algorithms. The blockchain provides a secure way to manage user registrations and logins without any central authority and Machine Learning models provide analytical prediction and intelligent detection of fraudulent activities. The method is based on transparency, reliability and intelligence.

Step 1: Smart Contract Development:

A smart contract using Solidity is created to authenticate users and help with the handling of data securely. The contract keeps secure all of a user's information on the chain e.g. (username, password hashed to protect the account). The contract has added functionality to store/retrieve the cid of content from the interplanetary file system, allowing for decentralized storage. The smart contract has defined events that log actions for users, including user registration/login/CID stored. The smart contract is deployed onto a local Ethereum blockchain using Ganache for the purpose of testing and development.

Step 2 Frontend Development

The frontend of this system was developed in React.js, making it user-friendly. This allows for users to register, log into their accounts to see if they have had any fraudulent activity, and monitor their transactions via the interface. There is a dashboard within the interface where users can see results of fraud detection and statistics of their transactions occurring in real-time. MetaMask connects actions performed by users in the application (logging in or registering) directly to blockchain transactions. The interface provides feedback on how the system functions and presents clarity around system operation to all users.

Step 3 :Web3 Integration

Web3.js lets the React frontend talk to the Ethereum blockchain. Web3.js lets people connect to an Ethereum network through MetaMask and use a smart contract that has already been deployed. All transactions, like making an account or storing data, will be safely signed and sent to the Ethereum blockchain. Web3.js asks the Ethereum blockchain for the smart contract function responses and then updates the frontend to match.

Step 4: Local Blockchain Setup

The Ganache Ethereum Blockchain Emulator provides a local environment for developing and testing Ethereum-based applications. It creates multiple test accounts that each hold a certain amount of virtual Ether for transacting on the blockchain. The Ganache environment makes it easy to deploy, debug, and monitor interactions with smart contracts. Ganache will validate all activity on the blockchain, including executing transactions and logging events

Step 5 Machine Learning Integration

Multiple ML models were trained using a financial fraud detection data set. The ML Model Types assigned for training include Logistic Regression, Random Forest, Gradient Boosting, XGBoost and LightGBM. Data prep processes were conducted on the original dataset to create a preprocessed and balanced training dataset to enhance model performance. Using evaluation metrics of performance, the models were assessed by accuracy, precision, recall and ROC-AUC score. The model selected as the best performer will be based on the model's performance from the evaluation. The best performing model was saved as a Joblib object and then utilized to predict in the back-end system.

Step 6 Decision Engine Implementation

To achieve a more accurate approach to classifying data, we will introduce a decision engine. Rather than simply taking the model output of the probability scores, we will process these as a threshold. By using the established threshold to classify the transactions as either fraud or non-fraud. Doing this will create a consistent means of making decisions on all requests while eliminating or reducing ambiguity in predictions.

Step 7 Flask Backend API

A backend developed with Flask manages communication to/from front end user interface and machine learning algorithms. Fraud detection and IPFS functionality have been coded into API endpoints and create transactions that can be processed on the backend. Transactions that are uploaded to the backend from the front end are processed by the backend transaction processing service which interacts with a trained model that can make predictions based upon the data. Then the backend sends the prediction results back to the front end for display on the dashboard. Additionally, the feature to generate the ROC curve date as well as to validate stored CIDs has been developed.

Step 8 Step 8: IPFS Integration

IPFS (InterPlanetary File System) enables files to be stored in a distributed manner. Rather than adding only the data from one document, the blockchain will contain only a hash of that document (CID). The document will have its actual file securely stored on IPFS, therefore, there will be less of a load on the blockchain. The system ensures the integrity of the data that is stored on the blockchain by checking the stored CIDs against what is recorded on the blockchain.

Step 9 Real-Time Dashboard Implementation

To create a real-time dashboard that will allow the visualization of transaction data and fraud detection results, the dashboard will include key metrics including the total number of transactions, the number of fraudulent transactions, the number of safe transactions, and the rate of frauds, as well as visual tools (slowly, steadily, pie chart, etc.) for visualizing the data. The dashboard will provide you with dynamic updates based on the processing of new transactions so that you can see them almost immediately after they have been completed. In order to ensure that you can monitor and analyze the data effectively, there are also options for alerts and exporting.

Step 10 Security and Validation

There will be input validation completed at both the front end and the smart contract sides of the application. Passwords will be securely hashed using keccak256 before they are saved into the blockchain. Only authenticated users will be able to access any system functionalities. The blockchain provides data immutability and prevents unauthorised changes/modifications to data on the blockchain.

Step 11: Testing and Deployment

Ganche and Truffle are both used to conduct smart contract testing as part of the Ganache testing environment. Each of the three components in the framework (the user interface; the server side; and the AI model) has been subjected to tests to validate their functionality. Several test cases were run to test both (1) the level of fraud detection accuracy produced, and (2) the level of system performance produced. For this project, all components have been developed, tested, and deployed using a “local sandbox” platform. There is the potential for future development to extend the solution onto public blockchain networks.

3.2 System Architecture Diagram

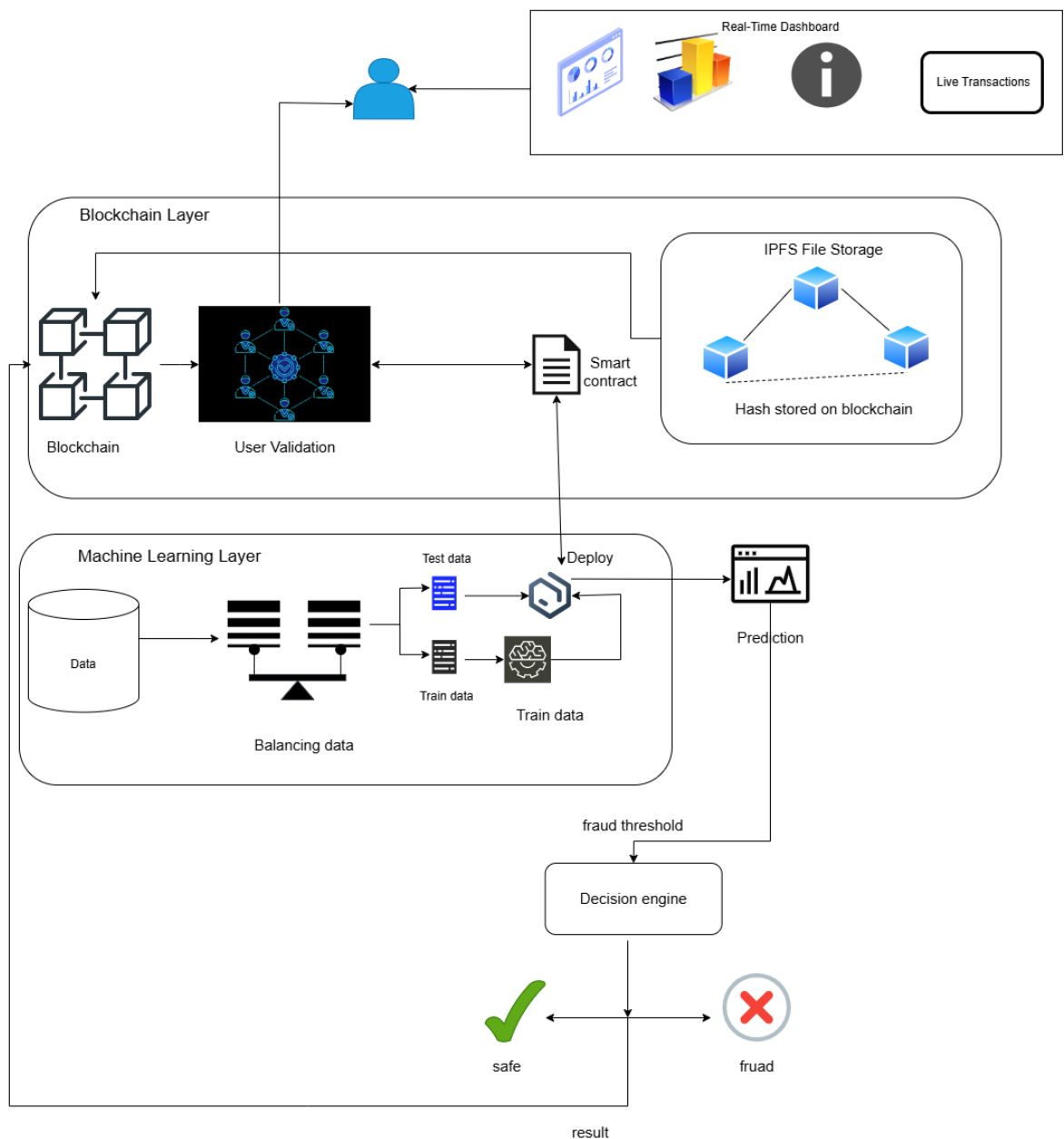


Figure 3.2.1 Architecture Diagram

The architecture diagram for your project can be described as follows:

1. **Blockchain:** The layer for storing user related information securely and in a decentralized way, keeping track of transactions without altering them and guaranteeing the transparency and trustworthiness of the system. In blockchain flagged user gets registered in blockchain and stays, no change is performed

to user unless admin does the change.

2. **Authenticating User:** Is represented as a group of nodes in a peer-to-peer relationship. The process of user authentication happens using block-chain technology. When user interface requests an information, the authentication of user is confirmed with decentralized validation.
3. **Smart Contract:** Smart contracts are to automatically execute actions required for operation of the system such as authenticating the users, updating fraud status of the users and controlling access. It also helps to store significant events in the block chain in a secured manner.
4. **Data:** Displayed by an database icon, which is the raw data to be used for training and testing the machine learning model. For example it can consist of past data, input by users or recorded by a blockchain transactions relating to the prediction required.
5. **Balance data:** This step tackle the class imbalance problem. Methods such as use of SMOTE. The reason why those are applied is to eliminate bias to the majority class.
6. **Training Data:** Post the balancing, the data is split into training data and testing data. The training subset is then fed Description of Algorithms
7. **Test Data:** A part of balanced dataset, reserved for testing the performance of model. It helps to check the model performance on unknown data.
8. **Deploy:** Illustrated by the hexagon, this stage is about taking the trained model and "deploying" it into the production environment. Now the model can begin to predict on new data, and may or may not be deployed into the system via the smart contract in a production setting.
9. **Predictions:** The ultimate output is an icon, which corresponds to a chart and represents the model's prediction. The prediction may be a class label, regression value, or probability depending on the application.
10. **IPFS Storage:** IPFS (InterPlanetary File System) is used for storing the files decentralized. Data is encrypted prior to upload for security purposes and only the file hash (CID) is stored in the blockchain, not the actual file, to

reduce on chain storage and improve efficiency.

11. **Decision Engine:** The prediction output will feed into a decision engine. It uses a specific threshold value to determine whether the transaction is "Fraud" or "Safe." This step makes the final result more definitive and reliable.
12. **Real-time Dashboard:** It has a real-time dashboard where you can observe live transactions, predicted values and analysis. This helps you to instantly monitor the system's activities and recognize fraudulent behavior immediately. The dashboard refreshes every 3 seconds.

3.3 Description of Algorithms

3.3.1 Random Forest

A Random Forest is a type of machine learning that improves decision trees' prediction power. Instead of relying on a single model it trains many decision trees and merges them to provide the answer to the problem. If a problem involves classifying then the answers of all the trees are voted on to provide the solution and if it involves prediction it provides the average answer from the trees. For this project we will be using a Random Forest in our attempt to detect fraud as they perform very well on data sets that have many variables and they are quite good at dealing with the variables. It is one of the models that we will be using for detecting fraud.

Key Characteristics:

1. **Avoid Overfitting:** Random Forest avoids over-fitting of the data by using more than a single tree, thus resulting in good accuracy on the data not encountered before.
2. **Feature importance:** Gives us details about how significant the variables used for fraud detection are.
3. **Ability to learn non-linear data:** The algorithm is able to discover non-linear relationships in data and therefore would be adequate for the fraud detection.

4. **Robustness:** It performs well even when the data contains noise or missing values, making it reliable for real-world applications.
5. **Parallel Processing:** Multiple trees can be trained independently, which improves computational efficiency.

3.3.2 XGBoost

XGBoost is a gradient boosting based advanced ensemble learning algorithm. It uses multiple decision trees in a sequence where new trees are created for correcting errors committed by the previous trees. It has gained a large popularity due to its great efficiency.

Key Characteristics:

1. **High Accuracy:** XGBoost provides excellent prediction accuracy due to its optimized boosting approach.
- 2 **Regularization:** It includes built-in regularization techniques that help prevent overfitting and improve generalization.
- 3 **Handling Missing Values:** XGBoost can automatically handle missing data, making it suitable for real-world datasets.
- 4 **Efficiency:** It is highly optimized for speed and can handle large datasets efficiently.

3.3.3 LightGBM

LightGBM is a fast and efficient gradient boosting algorithm that uses a leaf-wise tree growth strategy. It is designed for high performance and lower memory usage.

Key Characteristics:

- 1.**Faster Training:** LightGBM trains faster compared to traditional boosting algorithms due to its optimized structure.
- 2.**Memory Efficient:**
It uses less memory, making it suitable for large-scale data processing.

3 . Leaf-wise Growth: Instead of growing trees level-wise, it grows leaf-wise, which improves accuracy.

4. Handling Large Data: It performs well on large datasets with high dimensional features.

3.1.4 Gradient Boosting

Gradient Boosting is an ensemble technique that builds models sequentially, where each model tries to correct the errors made by the previous one. It combines weak learners (usually decision trees) to form a strong predictive model.

Key Characteristics:

1. Improved Accuracy: By correcting previous errors, it improves overall model performance.

2. Flexibility: It is possible to apply flexibility to classification and regression tasks, while capturing the intricate connections found among data points. In the sequential learning process, each model is dependent upon previous models; thus, this learning occurs in small increments which provide adequate precision.

3.1.5 Logistic Regression

Logistic Regression is a supervised learning algorithm used for classification tasks. It predicts the probability of a binary outcome using a logistic (sigmoid) function.

Key Characteristics:

1. Simple and Efficient: It is easy to implement and computationally efficient.

2. Probability Output: It provides probability values, which are useful for threshold-based decision making.

3. Interpretable Model: The model is easy to understand and explain compared to complex algorithms.

4. Baseline Model: It is often used as a baseline for comparing performance with advanced models.

3.4 Dataset Description

This dataset presents transactions that occurred in two days, where we have 492 frauds out of 284,807 transactions. The dataset is highly unbalanced, the positive class (frauds) account for 0.172% of all transactions. It contains only numerical input variables which are the result of a PCA transformation. Unfortunately, due to confidentiality issues, we cannot provide the original features and more background information about the data. Features V1, V2, ... V28 are the principal components obtained with PCA, the only features which have not been transformed with PCA are 'Time' and 'Amount'. Feature 'Time' contains the seconds elapsed between each transaction and the first transaction in the dataset. The feature 'Amount' is the transaction Amount, this feature can be used for example-dependant cost-sensitive learning. Feature 'Class' is the response variable and it takes value 1 in case of fraud and 0 otherwise. The second dataset is a financial fraud detection dataset obtained from Kaggle. This dataset is more detailed and is used for training and evaluating all machine learning models in this project. The dataset is organized into multiple categories, each representing different aspects of financial transactions:

- 1. Transaction Data:** The information given to characterize a transaction: the transaction id, transaction date/time, amount of the transaction (dollar amount), and customer information.
- 2. Customer Information:** Information about customer account profile, history, and transactions in chronologic order.
- 3. Fraud Indicators:** Information about unusual patterns and trends of suspicious transactions.
- 4. Transaction Amount Details:** Information about each transaction's amount, and indication on whether it's an outlier, using an anomaly score.
- 5. Merchant Information:** Information about merchant and transaction types, providing a financial system representative that matches the real system's complexity involved in one transaction. The goal of this dataset is to observe interactions between the customer's behavior, transaction trends and fraud detection, forming more complex relationships.

3.3.4 Requirement Specification

This project is the combination of blockchain technology and machine learning, to give a decentralized and smart system to perform identity verification and fraud detection. The system requirement specifications can be classified into functional and non-functional.

1. **User Registration:** The users should be able to register their username and password that would be stored immutably on the blockchain using the smart contract.
2. **User Login Verification:** The login credentials of the users have to be verified with that on the blockchain and accordingly login the users or reject the login attempt.
3. **Fraud Detection:** There should be a machine learning model used that should take the transactions made by users, and should detect fraudulence.
4. **Transaction Logging:** All the logins or transactions of users have to be logged on the blockchain immutably to provide traceability.
5. **MetaMask Integration:** users should have the ability to connect their Ethereum wallet in order to use the smart contract on Ganache or a test net.
6. **Decision Engine:** The system should apply a predefined threshold on the model output to classify transactions as “Fraud” or “Safe”.
7. **Real-Time Monitoring Dashboard:** The system should provide a dashboard to display live transaction data, fraud statistics, and analysis results
8. **Alert mechanism:** The system should generate alerts when a suspicious or fraudulent transaction is detected.
9. **Blockchain Integration :** Smart contracts should manage user data, authentication, and fraud status updates in a secure and immutable way.
10. **IPFS file storage:** The system should support secure file storage using IPFS, where files are encrypted and only their hash is stored on the blockchain.

3.5 System Workflow

The system workflow represents a series of actions for processing data, detecting fraud, and storing it. The process starts at the user end and goes through following all the way

to the output generation. Once you've written a sentence, output the whole revised paragraph so far. Through the app, the user can register or login, the authentication is done through blockchain with smart contracts. MetaMask is the wallet used to connect the user's wallet and sign transactions. The user may then submit transaction data to the system or perform operations. The backend preprocesses the input data. Such as the cleaning of the data, normalizing the data and using data augmentation for imbalanced data to make more balanced data. Then these processed data are sent to several machine learning models, Logistic Regression, Random Forest, Gradient Boosting, XGBoost and LightGBM. Every model starts out by analyzing the data. Based on the performance evaluation, the best one is chosen as the final prediction. The chosen model produces a probability score of being fraudulent. This output is fed into the decision engine which compares it against a specified threshold to determine if it is a fraud or not. The final result is exhibited in the real-time dashboard where users experience the complete transaction details, fraud status, and analytics. The system raises an alert if a fraudulent transaction is identified. Important information like transaction history and file identifiers are also saved securely with blockchain and IPFS. Blockchain guarantees the integrity of the data, and IPFS provides decentralized storage

3.6 Tools and Technologies Used

Recommended solutions and technologies that enable the development of machine learning, blockchain integration and web applications. The system's frontend is built with React.js and offers a dynamic and responsive UI. It provides users with a web-based dashboard with registration, log-in, and transaction monitoring functionality. React guarantees a quick rendering of the components, across the application. Ethereum is employed for blockchain integration, with smart contracts developed in Solidity. These user-authentication and data encryption processes are facilitated by the smart contracts. Ganache serves as the local blockchain environment to test and deploy smart contracts and MetaMask is used as a wallet to connect users to the blockchain and sign transactions. Web3.js is used to establish connection between the frontend and the blockchain, and interact with the smart contracts and execute transactions. Flask is used to develop the system's backend, which is a lightweight Python web framework. A temporary third-party service is employed to process API requests, q Integrate ML to Detect Anomalies in Transactions in osigyn. The following are some of the fraud

detection systems we've implemented via machine learning: Logistic Regression, Random Forest, Gradient Boosting, XGBoost, and LightGBM. These models can be trained and tested to identify the fraudulent transactions. The application makes use of IPFS (InterPlanetary File System) to store files in a decentralized way. It securely stores data and gives a content identifier (CID) which is logged in the blockchain as proof. Additional instruments, e.g., Joblib, are applied for saving trained models, and visualization libraries to generate graphs such as ROC curves and dashboards.

3.7 Summary

In this chapter, the proposed system design and its implementation are described in detail. The system architecture and design methodology describe the integration of a number of elements including machine learning agents, the user interface, blockchain and IPFS, to produce a full fraud detection system. Dataset observation and preprocessing The necessity for processing data prior to training model is well-understood Various. The system also describes multiple machine learning algorithms employed in the system to bring clarity on how each contributed towards detecting fraud transactions. The system flowchart explains how to clearly see the information flowing through the process at different levels, starting with the user interface level through to the final prediction to secure storage. The development process leverages flex values, cursor animations and pan gestures. To sum up, the chapter offers a detailed and holistic view of the suggested system design and construction which serves as building blocks for outcomes and analysis in the subsequent chapter.

CHAPTER-4

RESULTS AND OBSERVATIONS

A proposed machine learning system can quickly detect fake transactions for companies to focus their efforts on them. The project consists of a few steps, the first being to prepare the data. Here we need to clean the dataset and handle missing data and class imbalance techniques to ensure that all classes have an equal opportunity to build the machine learning model. Then some other special techniques can be implemented to improve the model performance. Then we need to train machine learning models like XGBoost, LightGBM, Random Forest, Gradient Boosting, Logistic Regression, etc. and evaluate it with regards to accuracy and ROC-AUC. In the next step, we need to select the best model which detects the fake transactions with better measures. Next, the model will create the transaction risk score where 'higher the score, higher the chances of transaction being fake' and we can have a decision engine that can take that score and compare against threshold and tell if the transaction is fake or safe. We need a dashboard that shows the transaction details and analytics for those details, where user can have an option to see the transaction history. An alert can be sent as an email or a text where the admin can assess those transactions as soon as possible. It's also recommended to log all the transactions which can be reviewed later and also helpful for the user to track their transaction.

4.1 Stepwise Description of Results

Fraud detection is a critical task for any organization dealing with user transactions. In today's tech-driven world, using Machine learning may provide solutions to efficiently detect fraud with high accuracy. This project is a Cloud-based Fraud Detection system powered using Machine learning and Blockchain which automatically detects fraudulent transactions using our machine learning models and sends instant notifications about the detected cases to the admin users at a centralized location. Working: The proposed system involves the following main stages. Dataset collection and preprocessing. The process starts with the collection of datasets containing information related to users

transactions. This dataset consists of a total of 0.5M records and consists of different features such as User ID, Transaction ID, type, Date-time, Amount, and Class, where class 1 indicates a fraudulent transaction and class 0 indicates a non-fraudulent transaction. This dataset is then preprocessed into the required format with a specified amount of missing values, normalization, and class balancing for fraudulent and non-fraudulent transactions.

ML Training and Evaluation The preprocessed dataset is then trained with various machine learning models to identify the best-suited model for getting high accuracy on a test dataset with more accuracy and less false prediction. The following ML models are used in this research paper, XGBoost, LightGBM, Random Forest, Gradient Boosting, Logistic Regression. The best model is then evaluated based on Accuracy and ROC-AUC metrics and once identified the best-performing model is deployed in the Machine Learning API endpoint for real-time predictions.

Real-time Fraud Detection System. The deployed machine learning model is then integrated with an Alert notification System in the system such that whenever a new transaction is generated the model scores the transaction with a confidence score based on its training and predicts a high likelihood of fraud. Based on whether the predicted fraud enabled it sends notifications and details regarding the transaction to the admin users at the centralized console. First it generates a detailed monitoring dashboard that includes transaction details along with fraud statistics and trends for required time intervals. Then an alerts section is provided that sends alerts on fraud detection to the user. A modeling section is provided to the user that shows real-time transaction logs with the respective prediction results and timestamps.

Blockchain/IPFS based secure storage The major issue of the transaction that focuses on the project is the security of user authentication as well as integrity of transaction record. Hence user's transaction data are shared to the various machines at a decentralised level but all transactions are maintained by the root admin. Thus we follow the Blockchain concept in our project. The IPFS is then used for maintaining the decentralised file storage. Thus before the original transaction file is stored in the IPFS the file is first encrypted using the RSA algorithm. Once encrypted the hashed file is stored in the IPFS. The file hash stored in IPFS is then kept on the Blockchain.

4.2 Result Analysis and Test Case Results

The performance of the proposed system is evaluated by applying multiple machine learning models on the dataset and comparing their results. Models such as Logistic Regression, Random Forest, Gradient Boosting, XGBoost, and LightGBM are trained and tested to identify the most suitable model for fraud detection and Live monitoring dashboard.

4.2.1 Detailed Analysis of Models

Logistic Regression is simple to use and effective for classification problems. It is useful for classification tasks. However, Logistic Regression is not good at identifying patterns in fraud. Random Forest is better due to the use of decision trees in combination. This helps the algorithm to cut down on errors and produce stable results. Gradient Boosting creates a model, which is even better, by trying to correct the errors, it made previously. Models like XGBoost and LightGBM perform even better. They produce results, as they have to learn smarter. They also train faster Can handle very large amounts of data. These models are good, for fraud detection real-world systems.

4.2.2 Mini Project vs Major Project Comparison

A comparison between the mini project and the major project is performed to highlight the improvements achieved in the system. The mini project primarily focused on basic fraud detection with a single model, combined with blockchain-based user verification and IPFS storage. Instead, the mini project was focused on basic fraud detection with a single model, blockchain-based user verification, and IPFS storage. The major project builds upon this by adding multiple machine learning models and comparing the results. The decision engine addition enhances decision-making by turning probabilities into classifications. There is also a real-time dashboard for visualization of results, an alert feature, and logging to enhance system monitoring. The system is more efficient, accurate, and user-friendly overall.

4.2.3 Test Case Results

The system is tested using different scenarios to verify its functionality and correctness. The test case results show that the system behaves correctly in various scenarios. It also blocks legitimate transactions. It also verifies the users' authentication via blockchain and produces an alert if suspicious activity is detected.

4.2.4. Machine learning models evaluation

The proposed fraud detection system has been evaluated by applying several machine learning models on the dataset and comparing their performance. The models under consideration in this work are Logistic Regression, Random Forest, Gradient Boosting, XGBoost and LightGBM. To ensure fairness in comparison, each model is run under identical conditions for training and testing. The assessment is performed through accuracy error, precision, recall, F1-score, ROC-AUC score. These statistics provide insight into each model's ability to detect fraudulent and legitimate transactions effectively. Among the models, logistic Regression achieves a compromised result for it is a simple linear model and can not capture complicated relationship of the data. Better result is observed through Random Forest, as it is ensemble of Decision Tree, in which overfitting can be reduced and stability of the model also improved. Gradient Boosting takes performance one step further by repeatedly correcting errors generated by previous models. Nonetheless, state-of-the-art boosting methods (XGBoost and LightGBM) achieve the best performance among all models. These models present a better accuracy and ROC-AUC, which indicates that they are more discriminating between fraudulent and non-fraudulent transactions. In particular, LightGBM is claimed to have faster training speed and higher efficiency, which is well-suited to large-scale datasets. The comparison highlights that the ensemble and boosting models perform better than the traditional models in all fraud detection tasks. According to the findings, the chosen model with the best performance is further analysed by confusion matrix and ROC curve. To summarize, employing multiple models in a complementary manner facilitates the selection of a robust technique and

thus makes the system more reliable and efficient.

Table 4.2.4 Machine learning model evaluation

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.586	0.583	0.601	0.592
Gradient Boosting	0.942	0.925	0.961	0.943
LightGBM	0.959	0.958	0.961	0.959
XGBoost	0.954	0.948	0.961	0.955

4.2.5 Confusion Matrix

A confusion matrix is a table used to evaluate the performance of a classification model. It shows the count of correct and incorrect predictions, categorized as:

1. **True Positives (TP):** Correct positive predictions
2. **True Negatives (TN):** Correct negative predictions
3. **False Positives (FP):** Incorrectly predicted positives
4. **False Negatives (FN):** Incorrectly predicted negatives

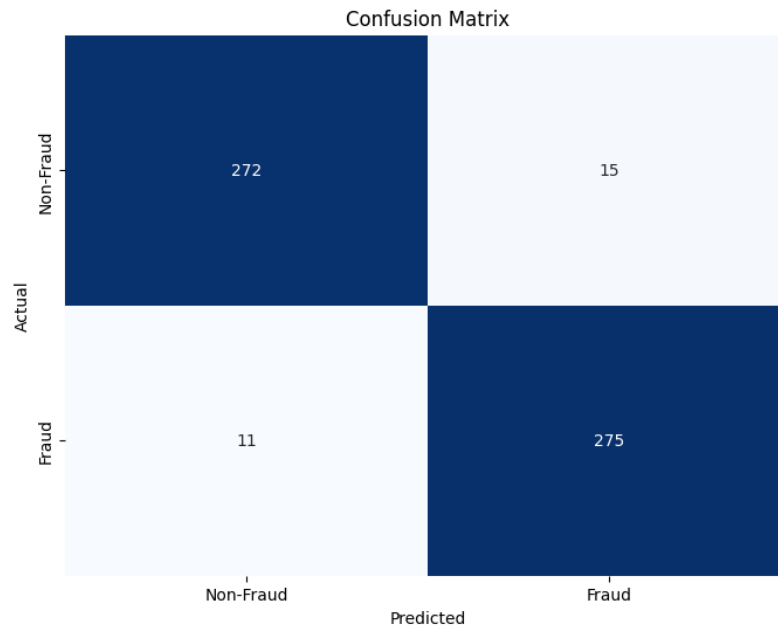


Figure 4.2.5 Confusion matrix

4.2.6 Precision -recall curve

1. Title: Precision-Recall Curve
2. X-Axis (Recall): Measures the proportion of actual positives correctly identified ($\text{True Positives} / (\text{True Positives} + \text{False Negatives})$). It ranges from 0 to 1.
3. Y-Axis (Precision): Measures the proportion of predicted positives that are actually correct ($\text{True Positives} / (\text{True Positives} + \text{False Positives})$). It also ranges from 0 to 1.
4. Curve Shape: The curve starts at a high precision (close to 1.0) with low recall (near 0), maintains high precision for a significant portion of recall values, and then drops sharply as recall approaches 1.0

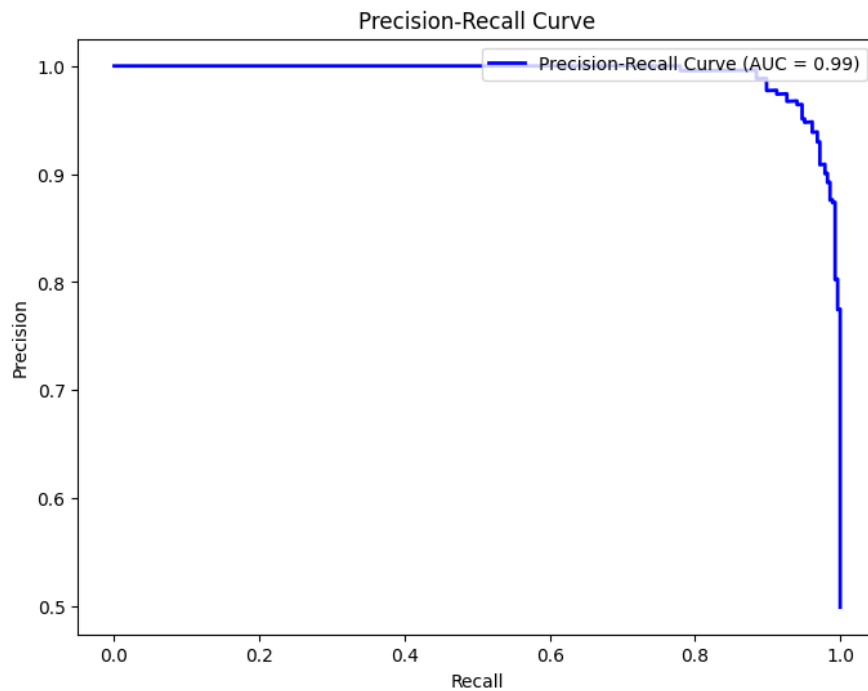


Figure 4.2.6 precision -Recall curve

4.2.7 Classification Report Metrics Visualization:

1. Chart Name: Classification Report Metrics Visualization

2. Classes: 0 and 1

3. Metrics:

Class 0: Precision = 1.0, Recall = 1.0, F1-Score = 1.0 (perfect)

Class 1: Precision = ~0.85, Recall = ~0.80, F1-Score = ~0.82
(good)

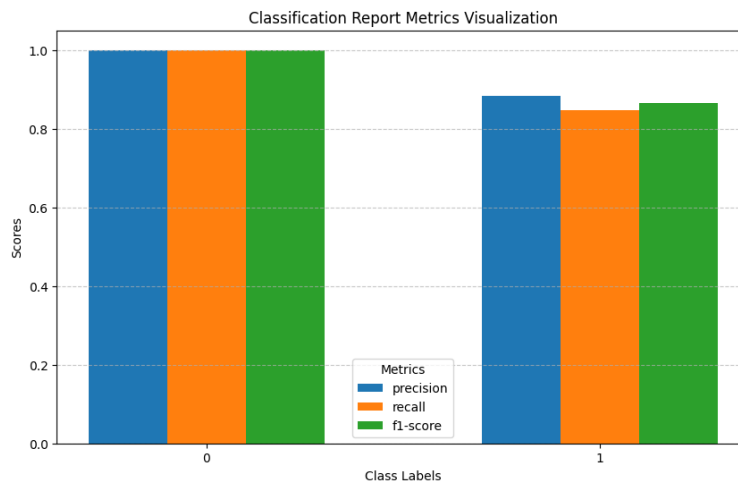


Figure 4.2.7 classification report

4.2.8 ROC CURVE

Compared to a random guessing process, the ROC curve illustrates how well the Classification system can detect fraudulent transaction activity by comparing its ability to identify true positives (or Sensitivity) and false alarms (Specificity) over a multitude of thresholds. The ROC curve comes close to the top left corner, indicating that many of the actual fraudulent transactions were successfully identified, and that the false alarm rate was low relative to the actual positives identified by the model. Thus, the classification system can accurately be said to detect fraudulent transactions with few mistakes. The dotted diagonal line represents what is expected if the classification model is not able to discriminate normal from fraudulent transactions; namely a random guessing scenario. The ROC curve coming in above this line shows that the classification system has produced a result significantly better than a random guess would produce. The AUC (Area Under the Curve) for this classification system is approximately 0.99; AUC of 1.0 is perfect; therefore the classification system nearly perfectly differentiates between legitimate and fraudulent transaction activity. To conclude, the ROC curve for this classification system indicates that it has the reliability and effectiveness necessary to be used in real-world applications for detecting fraudulent transaction activity.

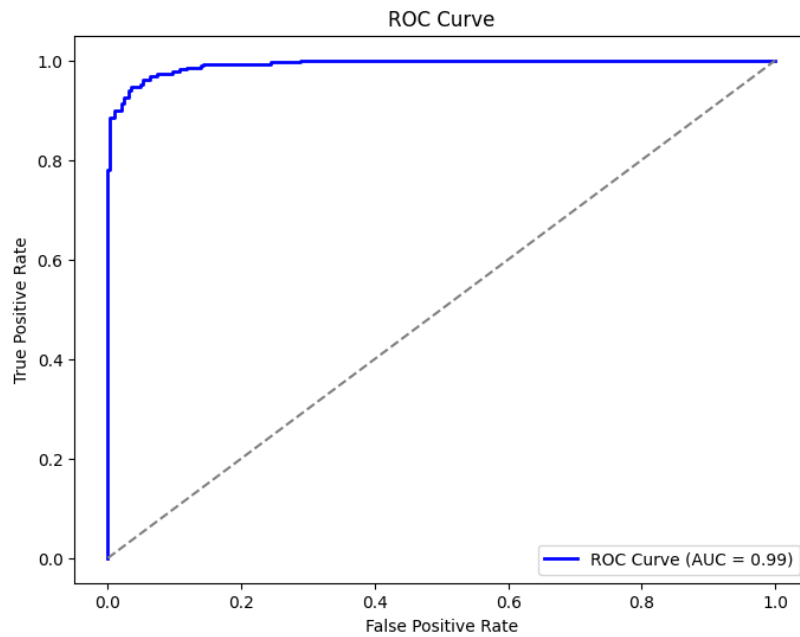


Figure 4.2.8 ROC curve

4.2.9 User Interface

The above figure shows the user interface for the developed decentralized application (DApp) that uses machine learning to detect fraud, and also to maintain the record of users securely on the blockchain. The UI has been developed with a view that it should be very simple and easy for users to use. The UI have the option of Registration, Login and Check fraud status. Before doing any action on this application, user has to connect his/her blockchain wallet, if wallet is not connected, then user is requested to "Connect to Metamask". The UI displays the "account connected", "fraud status", "Smart contract info". Buttons for "Switch Account" and "Check Fraud Status" gives additional options to user.

User Login/Register DApp

Account: 0xDD1f77E03957df5Bc0Fb11E15A1E9EDeA8Ab60da
Fraud Status: Legitimate
Contract Address: 0xC084Ea9a4eC52D7f8F87af1503ADB13E2b4b11d6
Registered: Yes

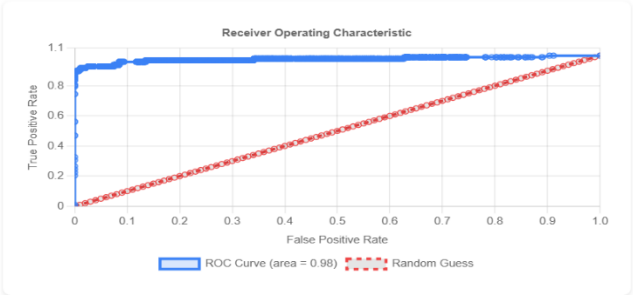
Switch Account Check Fraud Status

Username

Password

Register Login

This account is already registered.



Receiver Operating Characteristic

True Positive Rate

False Positive Rate

ROC Curve (area = 0.98) Random Guess

Figure 4.2.9 User interface

4.2.10 MetaMask Wallet Integration

Integration MetaMask wallet is integrated into the system to facilitate secure communication between the user and the blockchain. MetaMask serves as an intermediary between the user and the dApp and enables the user to link their Ethereum wallet accounts for performing transactions. During this connection process, the account details are obtained from the user's account and used as the method of authentication. Therefore, the conventional user and password authentication system is rendered useless in this system since all transactions and processes are performed using the blockchain system. Moreover, MetaMask provides some features such as transaction details like account balance, network status, and past transactions details. In this case, MetaMask can be configured for use on the local blockchain network (Ganache) to perform tests safely.

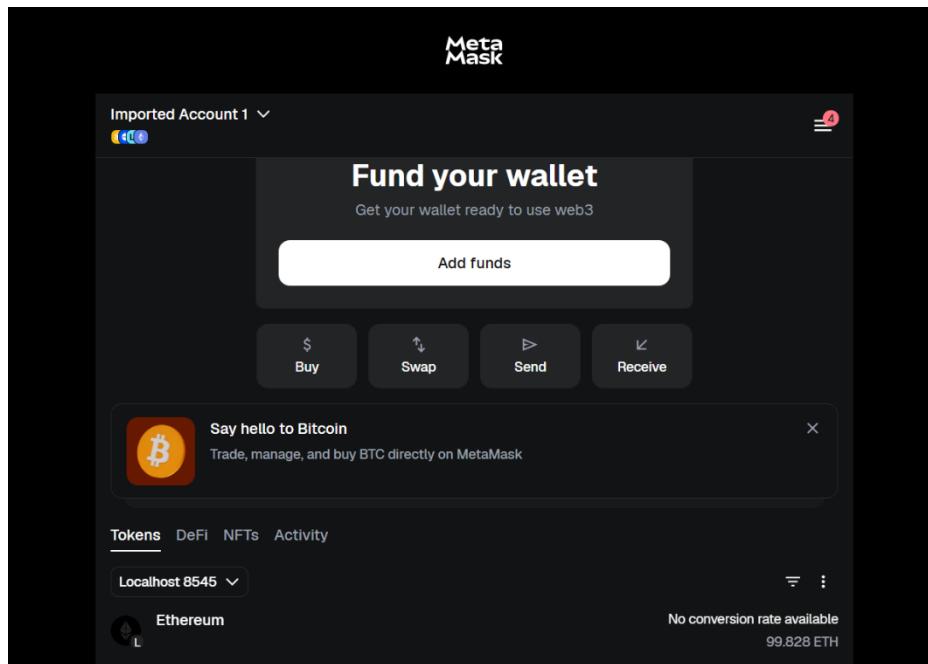


Figure 4.2.10 MetaMask integration

4.2.11 Blockchain Transaction

The above diagrams illustrate the process of executing transactions through blockchain through the use of Ganache, which is an Ethereum local testing environment. A transaction is executed every time there is any activity carried out by the user such as saving information or interacting with the smart contract. Each transaction will have specific details such as the transaction hash, the address of the sender, the address of the recipient (the contract), gas, and the block number. The details of the transaction help in establishing that the transaction was executed and stored on the blockchain successfully. It should be noted that the transactions done on the blockchain are immutable, meaning they cannot be changed or deleted. This helps in ensuring that there is transparency in the process.

4.2.12 Smart Contract Execution

The smart contract is another important component within this framework, which is responsible for verifying users and storing information. It will be

deployed on the blockchain and executes a particular set of actions whenever the user interacts with it. One of the most important components of this framework is the function called `storeCID()`. It allows us to store the content identifier from IPFS onto the blockchain. We will be able to save large amounts of data into the blockchain without any complications due to the use of this function. A smart contract is self-executing and does not involve any inputs or outputs by the user. The execution results of the smart contract can be traced by using the blockchain's log files. There is a transparent mechanism that allows all smart contract operations to be performed without complications. The use of a smart contract makes this framework much more reliable because it ensures that all the rules are enforced, and all the data is consistent with each transaction.

4.2.13 Backend API Integration

The application utilizes MetaMask, a cryptocurrency wallet, which connects a user to the Ethereum blockchain network. MetaMask provides a way for users to connect an Ethereum account to a decentralized application via the MetaMask bridge. MetaMask allows users to sign up for the application with a MetaMask account instead of using a traditional username and password authentication system. This improves security because only a user who has the corresponding private key can sign transactions. Along with signing transactions, MetaMask also provides users with other important information: account balance, connected network details, and history of transactions. In this case, MetaMask has been setup to connect with a local blockchain network (Ganache) making it safe to test and develop this integration of a user management application that is secure and decentralized.

4.2.14 Real-Time Fraud Monitoring Dashboard

The above figure illustrates the real-time fraud monitoring dashboard of the proposed system, which allows for live fraud detection and transaction activity monitoring via a dynamic, interactive graphical interface. The top section of the dashboard contains KPIs (key performance indicators) indicating the total number of transactions; the number of fraud, non-fraud, and total transactions;

and the overall rate of fraud. These KPIs provide a quick visualization of the current state of the system and an understanding of how transactions are distributed. In addition, the dashboard has control buttons for simulating transactions, pausing live transactions, refreshing immediately; exporting data as CSV, and going back to the upload module, each helpful for managing and analyzing transaction data. A pie chart, titled “Fraud vs Non-Fraud”, provides a graphic representation of fraud detection and clearly illustrates the proportion of fraudulent and non-fraudulent transactions. Currently, the pie chart indicates that all transactions are classified as non-fraudulent. The dashboard has a line chart, “Fraud Incidence Over Time”, enabling tracking of the incidence of fraud over time so that trends in the incidence of fraud can be identified, as can trends in transaction behavior. This is important for performing analysis and developing Utah State University’s future business strategy.

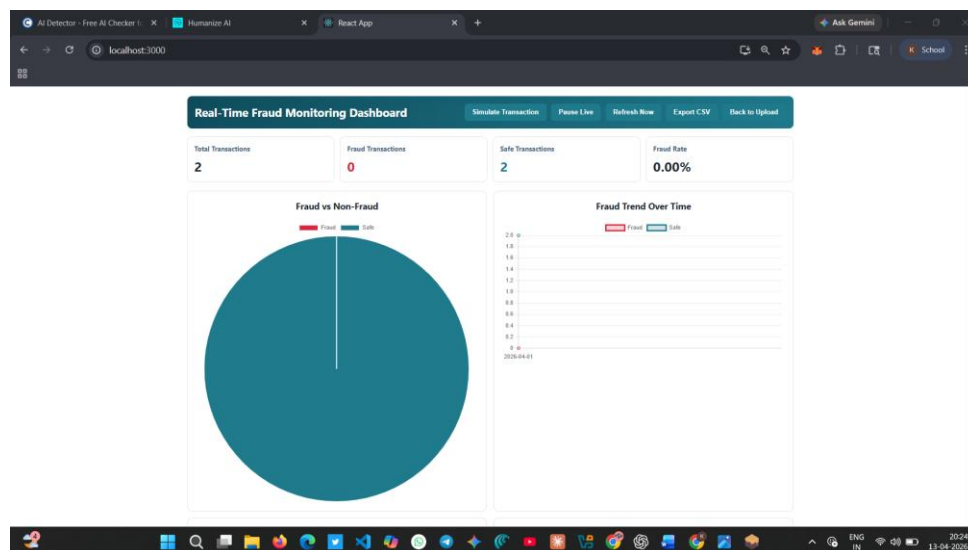


Figure 4.2.14 Real-Time monitoring dashboard

4.2.15 Overall Analysis

From the results, it is clear that the proposed system performs better than the traditional approaches. The proposed system incorporates multiple machine learning models, optimizing accuracy, and providing a basis for choosing the most appropriate models. The decision engine provides a clearer classification,

while the dashboard in real-time enhances the system's usability and provides a monitoring interface. The integration of blockchain and IPFS guarantees data security and integrity, which enhances the reliability of the system.

4.3 Observations from the Work

4.3.1 Performance of Machine Learning Models

The investigations highlighted various outcomes from individual machine learning models' capacity to predict fraudulent transactions: within the results of the various different models used; it was observed that there was also difference in performance levels observed across all types of models. Generally speaking, Logistic Regression model performance on this data was moderate, but it couldn't adequately model the overall complexity of patterns contained within multiple linear distribution forms. In addition; another class of strategies enabling superior model performance would be those based on an ensemble approach. Random Forest and Gradient Boosting are both examples of this class of predictive technique because they combine the results produced from multiple independent decision trees and prevent overfitting. When reviewing overall capabilities, the top performing models when using training data including advanced boosting models are XGBoost and LightGBM. Both XGBoost and LightGBM have demonstrated superior ability to provide accurate predictions with greater levels of precision and recall than other models utilized for fraud detection efforts. Additionally, both XGBoost and LightGBM have demonstrated the ability to effectively handle large amounts of training data using very complicated data relationships thereby providing a substantive advantage in their ability to be deployed in a wide range of applications.

4.3.2 Impact of Data Preprocessing and Balancing

An additional very important point to note is that pre-processing of data impacts performance on the fraud detection system. Fraud Detection using Imbalanced datasets is usually associated with the detection system used for detecting fraud. Without proper handling of these datasets, the model developed will classify the

vast majority of transactions as being normal transactions. The Fraud Detection System is based upon Supervised Learning and the performance of the Fraud Detection System will be enhanced after applying the datasets balancing techniques as outlined above. Additionally, there are other pre-processing steps (normalization, treatment and/or removal of the features) that should be performed in order to support the improvement of the model's accuracy in detecting cases of fraud. As a result, data mining processes such as dataset balancing, normalizing data, and handling of the data features all enhance the model accuracy with respect to performing fraud detection.

4.3.3 Role of Decision Engine

Using blockchain technology on the system increases the amount of security and transparency available to it. This also guarantees that both user data and transaction records are stored in a decentralized, uneditable manner. Hence, it is impossible to access or modify anyone's information in any unauthorized manner. Smart contracts will automate important processes such as authenticating users and updating their fraud status, which reduces the number of times these processes need to be performed manually, creating more dependable systems than traditional techniques, as well as improving trust and security of both the system and the users who use it.

4.3.4 Role of IPFS in Secure Storage

Decentralized storage of files relies heavily on IPFS. Instead of copying all data into a blockchain, the blockchain only maintains the file hash (CID) and uses IPFS for actual storage of the data. Consequently, this solution provides considerable space savings since many files will not be on the blockchain; thus the efficiency of all systems involved will improve. Moreover, the use of encryption increases the security of any sensitive information. The combination of IPFS and blockchain results in a highly secure, scalable solution for data storage and sharing, dramatically reducing the amount of space and effort associated with storing and sharing large amounts of information securely.

4.3.5 Effectiveness of Blockchain Integration

The combination of the blockchain technology disciplines provide an increase in security and transparency for the system. User data and transaction records are kept securely in a decentralized, unchanging manner to prevent unauthorized access and data tampering. Using smart contracts, many important functions are automated, such as user authentication and fraud tracking. Due to the automation of these functions, the amount of manual intervention decreases, which increases the overall reliability of the system. Therefore, compared to traditional systems, systems using blockchain technology offer greater levels of trust and security.

4.3.6 Importance of Real-Time Dashboard

The Real-time Dashboard enhance the system to be user-friendly and clear in showing the transactions data, fraud result and system performance. With the dashboard, the user can observe the currently going on transactions and any suspicious actions in the transactions clearly. It also shows some statistics such as fraud trend and transactions statistic so that the user can learn the system more about the system performance. This was something that was not present in the mini project and makes it a good new addition.

4.3.7 Alert System and Monitoring

If there's any indication of something suspicious, you'll immediately receive an alert from the system. If any suspicious activity is detected, an alert will be sent to your inbox or as an email. That notice can help people get things moving quickly to prevent additional damage from occurring. All activity is captured and recorded in log files for future reference. By analysing these patterns over time, the overall effectiveness of these systems will continue to improve.

4.3.8 Comparison with Mini Project

Major project improvements. Compared to mini project, major project makes great improvements in lots of fields. Mini project only contains one model and

only implement fraud detection function. Mini project did use block chain and IPFS but lacks functions such as model comparison, decision engine, and real-time monitor. Major project built upon mini project and implemented models of machine learning, better decision making and better visualization.

4.3.9 Overall System Performance

In general, the suggested system offers improved accuracy, security, and user friendliness compared to others. A complete solution for fraudulent activity detection has been created through the use of machine learning technology, in combination with blockchain technology, IPFS, and capability for real-time monitoring. The suggested system is able to process complex data, detect fraudulent activity effortlessly, and manage data securely.

4.4 Model Performance Comparison

The performance of the proposed fraud detection method is explored using multiple classification algorithms on the dataset. The classification algorithms used in this study are Logistic Regression (LR) Random Forest (RF) Gradient Boosting (GB) XGBoost and LightGBM. All algorithms have been trained and tested on the same ground to get a balanced and equal comparison between models. The performance of these algorithms is investigated using significance tests for accuracy precision recall F-score and ROC- AUC score. These statistics are used to measure the accuracy of each of the models in correctly classifying transactions as fraudulent or non-fraudulent. LR can provide average performance as its a simple linear model and may not represent non-linear relations. Random Forest can achieve better performance because it involves the ensemble technique which makes prediction stable because they use many trees. GB also enhances the performance as they iteratively compensate on the previous models errors. XGBoost and LightGBM achieve high performance on all the metrics used. More sophisticated boosting algorithms are achieved in higher accuracy and ROC-AUC scores. Both indicate the great capacity of models to capture the fraud patterns. Specifically, LightGBM is well-known for faster training and handling of large dataset. The comparison of these algorithms clearly suggests that boosting based algorithms perform better than classical

approaches on the problem of credit card fraud detection. The top model among the evaluated models has been selected and tested with a further comparison using confusion matrix and roc Curve based analysis. Generally comparing the models is a fundamental step for building the most suitable algorithm to boost system performance hence reducing the fraud transactions.

4.5 Summary

In this chapter, the proposed system results are discussed and analyzed in detail. Consequently, the following stepwise explanation illustrates how the system processes the data, along with the application of the ML model and the prediction outputs in the fraud detection. The flow of work depicts how the data preprocessing, model training, decision engine and secure storage modules are integrated. The comparison of the performance of various machine learning models is based on the evaluation measures which indicate that the sophisticated boosting models i.e. XGBoost and LightGBM have higher accuracy and run time efficiency. Result analysis also focuses the enhancement of new scheme over the previous work especially in model selection and prediction aspect. Simulation results verify the system's accurate detection of both fraud and normal transaction and integration with two-factor authentication. Selected model performance is also validated by confusion matrix and ROC. Some of the observations include insights into system behavior, such as how data preprocessing affected the results, how well the decision engine worked, and how blockchain and IPFS were used to protect data. The real-time dashboard enhances monitoring and feasibility through intuitive visualization of transaction data. In general, the chapter shows that the proposed scheme can be run effectively, with high precision in fraud detection and safe and feasible in practical use.

CHAPTER-5

CONCLUSION AND FUTURE WORK

5.1 Conclusion

In this project we created a system to detect fraud efficiently, which is made up of ML, Blockchain and IPFS. Since there are certain weaknesses in previous systems, we created one which is decentralized and not alterable so that the users are reliable and the information can be stored safely. We experimented with machine learning models such as: XGBoost, LightGBM, Random Forest, Gradient Boosting, and Logistic Regression to assess their effectiveness in detecting fraudulent activity, and decided to use a decision making engine to evaluate a transaction as fake or real. This method is helpful to clarify where we should decide if it is or is not a fake transaction. This blockchain technology makes our data secure, and each transaction can be publicly audited, the IPFS helps us store documents in a decentralized way. The application includes a live dashboard to view transaction and its status in real time, a system to log each transaction, an alert system and controls so that the relevant users are informed and can track any suspect users. We showed in this project that our system works well and it can manage high amounts of tasks and it's really good at identifying fraudulent transactions, which suggests our system can be applied to the world in terms of fraud detection, which the fraud detection system actually is; it is meant to detect fraud.

5.2 Future study

The proposed system can be improved in several aspects to increase its efficiency and effectiveness. In future the system can be implemented on a blockchain network instead of on a standalone machine to evaluate its performance in scalability, speed of transaction processing and cost. We can test more advanced machine learning and deep learning models on the proposed system to enhance its capabilities in identifying fraud and evolving with new patterns of fraud. The system can be implemented on time-based detection of data for an efficient, near real-time fraud identification in huge networks like that of banks and online purchasing stores. Security can be implemented to the system by requiring a username, password and authentication code for system access, restricting user access to specified functionalities, and ensuring the confidentiality of

data. The system can also be optimized to handle massive amounts of data and a huge number of transactions at a time, which would enable large corporations to utilize it effectively. Usability of the system can be improved, and the data inspection tools can be enhanced to help users make informed decisions. The described system can be implemented in the suggested ways, to make the system more efficient at detecting fraud and more convenient for users.

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Appendix

```
const initWeb3 = async () => {
  try {
    let web3;
    let account;

    // Check if MetaMask is available
    if (typeof window !== 'undefined' && typeof window.ethereum !== 'undefined') {
      web3 = new Web3(window.ethereum);

      try {
        // Request account access if needed
        const accounts = await web3.eth.getAccounts();
        if (accounts.length === 0) {
          await window.ethereum.request({ method: 'eth_requestAccounts' });
          const newAccounts = await web3.eth.getAccounts();
          account = newAccounts[0];
        } else {
          account = accounts[0];
        }
      } catch (error) {
        console.error("MetaMask connection error:", error);
        throw new Error("Failed to connect to MetaMask");
      }
    } else {
      // Fallback to Ganache
      web3 = new Web3('http://127.0.0.1:7545');
      console.log("Connected to Ganache");

      const accounts = await web3.eth.getAccounts();
      account = accounts[0];
    }
  }
}
```

Figure A.1: Blockchain connection to frontend

```
const loadBlockchain = async () => {
  try {
    const { web3, account, contract } = await initWeb3();
    setWeb3(web3);
    setAccount(account);
    setContract(contract);

    const user = await contract.methods.users(account).call();
    setIsRegistered(user.exists);
    setIsFraudulent(user.isFraudulent);
    setMessage(user.exists ? "This account is already registered." : "");
    if (user.isFraudulent) {
      setMessage("⚠️ This account is flagged as fraudulent!");
    }
  } catch (error) {
    console.error("❌ Failed to load web3:", error);
    setMessage(
      "❌ Failed to connect to blockchain. Please ensure MetaMask is installed and connected."
    );
  }
};
```

Figure A.2: User register in Blockchain

```

const checkFraud = async () => {
  if (!account) {
    setMessage("❌ Please connect MetaMask first.");
    return;
  }
  try {
    const response = await axios.post("http://localhost:5000/predict", {
      userAddress: account,
      transaction: buildDemoTransaction(),
    });
    if (response.data.prediction === 1) {
      setMessage(
        `⚠️ Fraud detected! Probability ${
          response.data.fraud_probability * 100
        }.toFixed(2)}%`,
      );
      alert("⚠️ Fraudulent Transaction Detected!");
      setIsFraudulent(true);
    } else {
      setMessage(
        `✅ Transaction is legitimate. Probability ${
          response.data.fraud_probability * 100
        }.toFixed(2)}%`,
      );
    }
    await fetchDashboardData();
  } catch (error) {
    console.error(
      "Fraud check error:",
      error.response?.data || error.message,
    );
    setMessage(
      `❌ Failed to check fraud status: ${
        error.response?.data?.error || error.message
      }`,
    );
  }
};

```

Figure A.3: Fraud Detection implmentation

```

// Store CID on blockchain
setMessage("📁 Storing CID on blockchain...");
await contract.methods.storeCID(cidString).send({
  from: account,
  gas: 200000,
  gasPrice: await web3.eth.getGasPrice(),
});
setMessage(`✅ CID stored on blockchain! CID: ${cidString}`);
await getCIDs();
} catch (error) {
  setMessage(`❌ Failed to upload file: ${error.message}`);
}
};

```

Figure A.4: File storage in CID

```

# Define model
xgb_model = XGBClassifier(
    objective='binary:logistic',
    eval_metric='logloss',
    random_state=42,
    use_label_encoder=False
)

# Hyperparameter tuning
param_grid = {
    'n_estimators': [100, 200],
    'max_depth': [4, 6, 8],
    'learning_rate': [0.01, 0.1],
    'subsample': [0.8, 1.0]
}

grid_search = GridSearchCV(
    estimator=xgb_model,
    param_grid=param_grid,
    cv=3,
    scoring='accuracy',
    n_jobs=-1
)

grid_search.fit(X_train, y_train)

best_model = grid_search.best_estimator_

```

Figure A.5: Code Snippet of XGB model

```

from lightgbm import LGBMClassifier

lgb_model = LGBMClassifier(
    objective='binary',
    random_state=42
)

param_grid = {
    'n_estimators': [100, 200],
    'max_depth': [-1, 10, 20],
    'learning_rate': [0.01, 0.1],
    'num_leaves': [31, 50]
}

grid_search = GridSearchCV(
    estimator=lgb_model,
    param_grid=param_grid,
    cv=3,
    scoring='accuracy',
    n_jobs=-1
)

grid_search.fit(X_train, y_train)

best_model = grid_search.best_estimator_

```

Figure A.6: Code Snippet of lightgbm

```

from sklearn.ensemble import GradientBoostingClassifier

gb_model = GradientBoostingClassifier(random_state=42)

param_grid = {
    'n_estimators': [100, 200],
    'learning_rate': [0.01, 0.1],
    'max_depth': [3, 5]
}

grid_search = GridSearchCV(
    estimator=gb_model,
    param_grid=param_grid,
    cv=3,
    scoring='accuracy',
    n_jobs=-1
)

grid_search.fit(X_train, y_train)

best_model = grid_search.best_estimator_

y_pred = best_model.predict(X_test)
y_pred_proba = best_model.predict_proba(X_test)[:, 1]

```

Figure A.7: Code Snippet of Gradient Boosting

```

from sklearn.linear_model import LogisticRegression

log_model = LogisticRegression(max_iter=1000)

param_grid = {
    'C': [0.1, 1, 10]
}

grid_search = GridSearchCV(
    estimator=log_model,
    param_grid=param_grid,
    cv=3,
    scoring='accuracy',
    n_jobs=-1
)

grid_search.fit(X_train, y_train)

best_model = grid_search.best_estimator_

y_pred = best_model.predict(X_test)
y_pred_proba = best_model.predict_proba(X_test)[:, 1]

```

Figure A.8 : Code Snippet of Logistic Regression

```
# Hyperparameter tuning for RandomForestClassifier
param_grid = {
    'n_estimators': [100, 200, 300],
    'max_depth': [None, 10, 20],
    'min_samples_split': [2, 5, 10],
    'min_samples_leaf': [1, 2, 4]
}
model = RandomForestClassifier(random_state=42)
grid_search = GridSearchCV(estimator=model, param_grid=param_grid, cv=3, scoring='accuracy', n_jobs=-1)
grid_search.fit(X_train, y_train)
```

Figure A.9 : Code Snippet of Random Forest

7% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Match Groups

-  **40 Not Cited or Quoted 6%**
Matches with neither in-text citation nor quotation marks
-  **0 Missing Quotations 0%**
Matches that are still very similar to source material
-  **7 Missing Citation 1%**
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

-  **6% Internet sources**
-  **4% Publications**
-  **0% Submitted works (Student Papers)**

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

13% detected as AI

The percentage indicates the combined amount of likely AI-generated text as well as likely AI-generated text that was also likely AI-paraphrased.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Detection Groups

- 21 AI-generated only 13%
Likely AI-generated text from a large-language model.
- 0 AI-generated text that was AI-paraphrased 0%
Likely AI-generated text that was likely revised using an AI-paraphrase tool or word spinner.

Disclaimer

Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?

The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?

Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

